

# P4410 Problem set 12 solutions

Shankar 19.3.1

From Shankar Eq. 19.3.11

$$\frac{d\sigma}{d\Omega} = \frac{4\mu^2 g^2}{\hbar^4 [\mu_0^2 + 4k^2 \sin^2(\theta/2)]^2}$$

$$\int d\sigma = \int \frac{4\mu^2 g^2}{\hbar^4 [\mu_0^2 + 4k^2 \sin^2(\theta/2)]^2} d\Omega. \quad d\Omega = \sin\theta d\theta d\phi$$

$$\sigma = \int_0^\pi \int_0^{2\pi} \frac{4\mu^2 g^2 \sin\theta}{\hbar^4 [\mu_0^2 + 4k^2 \sin^2(\theta/2)]^2} d\phi d\theta$$

$$= \frac{8\pi\mu^2 g^2}{\hbar^4} \int_0^\pi \frac{\sin\theta}{[\mu_0^2 + 4k^2 \sin^2(\theta/2)]^2} d\theta$$

Integration yields

$$= \frac{8\pi\mu^2 g^2}{\hbar^4} \left( \frac{2}{4k^2\mu_0^2 + \mu_0^4} \right)$$

$$= \frac{16\pi^2 g^2}{\hbar^4 \mu_0^2} \left( \frac{1}{\frac{4k^2}{\mu_0^2} + 1} \right)$$

substituting  $r_0 = 1/\mu_0$   
and noting  $r_0 \mu_0 = 1$

$$\sigma = \frac{16\pi^2 g^2 r_0^2}{\hbar^4} \left( \frac{1}{4k^2 r_0^2 + 1} \right) = 16\pi^2 r_0^2 \left( \frac{g\mu_0}{\hbar^2} \right)^2 \frac{1}{1 + 4k^2 r_0^2} \quad \checkmark$$

Shankar 19.3.2

$$(1) \quad f(\theta) = -\frac{2\mu}{\hbar^2} \int \frac{\sin qr'}{q} V(r') r' dr', \quad V(r) = -V_0 \Theta(r_0 - r)$$

localized to  
sphere of  
radius  $r_0$

$$= \frac{2\mu V_0}{\hbar^2 q} \int_0^{r_0} r' \sin qr' dr'$$

Integration yields

$$= \frac{2\mu V_0}{\hbar^2 q} \left[ -\frac{r_0 \cos qr_0}{q} + \frac{\sin qr_0}{q^2} \right]$$

$$f(\theta) = \frac{2\mu V_0}{\hbar^2 q} \left[ \frac{\sin(qr_0) - r_0 q \cos(qr_0)}{q^2} \right]$$

and  $\frac{d\sigma}{d\Omega} = |f(\theta)|^2$

$$\Rightarrow \frac{d\sigma}{d\Omega} = \left( \frac{\mu V_0}{\hbar^2 q^3} \right)^2 \left( \sin(qr_0) - r_0 q \cos(qr_0) \right)^2 \left( \frac{r_0}{q} \right)^6$$

$$\frac{d\sigma}{d\Omega} = 4r_0^2 \left( \frac{\mu V_0 r_0^2}{\hbar^2} \right)^2 \left[ \frac{(\sin(qr_0) - qr_0 \cos(qr_0))^2}{(qr_0)^6} \right] \checkmark$$

Shankar 19.3.2

(2) As  $k_0 \rightarrow 0$ ,  $q_0 \rightarrow 0$

We examine the factor in brackets from part (1)

$$\lim_{\epsilon \rightarrow 0} \frac{(\sin \epsilon - \epsilon \cos \epsilon)^2}{\epsilon^6} \quad \text{Taylor expand.}$$

$$\lim_{\epsilon \rightarrow 0} \frac{\left( \epsilon - \frac{\epsilon^3}{3!} + \dots - \epsilon \left( 1 - \frac{\epsilon^2}{2!} + \dots \right) \right)^2}{\epsilon^6} = \lim_{\epsilon \rightarrow 0} \left( \cancel{\epsilon} - \frac{\epsilon^3}{3!} + \dots - \cancel{\epsilon} + \frac{\epsilon^3}{2!} + \dots \right)^2$$

$$\lim_{\epsilon \rightarrow 0} \frac{(\epsilon^3(\frac{1}{2} - \frac{1}{6}))^2}{\epsilon^6} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon^6}{\epsilon^6} \frac{1}{9} \quad \text{Clearly (or applying L'Hopital's rule 6 times) yields } \frac{1}{9}.$$

Thus picking up  $4\pi$  from angle independent  $\phi$  we obtain

$$\nabla = \frac{16\pi f_0^2}{9} \left( \frac{4V_0 f_0^2}{\hbar^2} \right)^2 \quad \checkmark$$

Shankar 19.5.1 100 MeV neutron incident on nucleus

$$l_{\max} \approx kr_0$$

$$r_0 \approx 1F = 10^{-5} \text{Å}$$

$$k = \frac{\sqrt{2EM}}{\hbar}$$

$$E = 100 \text{ MeV}, \quad M = 938 \text{ MeV}$$

$$\hbar c \approx 200 \text{ MeV F}$$

$$l_{\max} \approx \frac{\sqrt{2 \cdot 100 \text{ MeV} \cdot 938 \text{ MeV}}}{200 \text{ MeV F}} 1F$$

$$\approx 2.17$$

or

$$l_{\max} \approx 2 \checkmark$$

④ Shankar 19.5.3

$$1) \lim_{kr_0 \rightarrow 0} \delta_l = (kr_0)^{2l+1}$$

For small  $kr_0$ , only need to consider  $l=0$   
 so  $\delta_0 \approx kr_0$

$$\sigma \approx \sigma_0 \approx \frac{4\pi}{k^2} \sin^2 \delta_0 \approx \frac{4\pi \delta_0^2}{k^2} \approx 4\pi r_0^2$$

Note that this is 4x larger than classical geometric cross-section.

$$2) \delta_l = \arctan \left[ \frac{j_l(kr_0)}{n_l(kr_0)} \right] \underset{kr_0 \rightarrow \infty}{\approx} \arctan \left[ \frac{\sin(kr_0 - l\frac{\pi}{2})/kr_0}{-\cos(kr_0 - l\frac{\pi}{2})/kr_0} \right]$$

$$= -(kr_0 - l\frac{\pi}{2})$$

$$\text{so } \sin^2 \delta_l \rightarrow \sin^2(kr_0 - l\frac{\pi}{2}) \quad \text{QED.}$$

⑤ For  $l=0$ , wave functions are spherically symmetric:  
 $\psi = R(r)$ . Use radial equation (Shankar 12.6.5):

$$\left\{ \frac{d^2}{dr^2} + \frac{2m}{\hbar^2} \left[ E - V(r) - \frac{l(l+1)\hbar^2}{2mr^2} \right] \right\} U(r) = 0$$

where  $U(r) = rR(r)$ , and  $l=0$ :

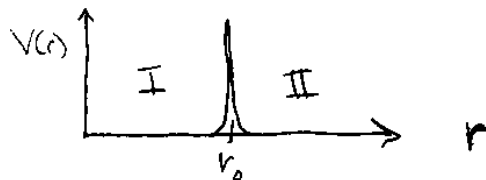
$$\frac{d^2 U}{dr^2} + \left( \frac{2mE}{\hbar^2} - \frac{2mV(r)}{\hbar^2} \right) U(r) = 0$$

now  $E = \frac{\hbar^2 k^2}{2m}$  so

$$\frac{d^2 U}{dr^2} + \left( k^2 - \frac{2mV(r)}{\hbar^2} \right) U(r) = 0$$

Solutions in  $V=0$  regions:  
 $U(r) = Ae^{ikr} + Be^{-ikr}$

Divide up regions radially:



Region I:  $U_I(r) = A'e^{ikr} + B'e^{-ikr}$  constraint  $A' = -B'$  due to  $U(0)=0$ .

so  $U_I = A \sin kr$ , redefining  $A = \frac{2A'}{i}$

$$U_{II}(r) = C e^{ikr} + D e^{-ikr}$$

Constraints on  $U_{II}(r)$ :

$$U_{II}(r_0) = U_I(r_0): C e^{ikr_0} + D e^{-ikr_0} = A \sin kr_0 \quad (*)$$

Constraints on  $\frac{dU}{dr}$  at  $r_0$ :

$$\int_{r_0-\epsilon}^{r_0+\epsilon} \left\{ \frac{d^2 U}{dr^2} + k^2 U(r) - \frac{2m\delta(r-r_0)}{\hbar^2} U(r) \right\} = 0$$

As on 1-D Schrödinger eqn problems, take  $E \rightarrow 0$ :

$$\left. \frac{dU_{II}}{dr} \right|_{r_0} - \left. \frac{dU_I}{dr} \right|_{r_0} - \frac{k^2}{0} U(r_0) - \frac{2ma}{\hbar^2} U(r_0) = 0.$$

$$iCk e^{ikr_0} - iDk e^{-ikr_0} - Ak \cos kr_0 = \frac{2ma}{\hbar^2} A \sin kr_0$$

Substitute  $A \rightarrow \frac{C e^{ikr_0} + D e^{-ikr_0}}{\sin kr_0}$  from (\*).

$$iCk e^{ikr_0} - iDk e^{-ikr_0} - k(C e^{ikr_0} + D e^{-ikr_0}) \cot kr_0 = \frac{2ma}{\hbar^2} (C e^{ikr_0} + D e^{-ikr_0})$$

$$C \left[ ike^{ikr_0} - ke^{ikr_0} \cot kr_0 - \frac{2ma}{\hbar^2} e^{ikr_0} \right] = D \left[ \frac{2ma}{\hbar^2} e^{-ikr_0} + ike^{-ikr_0} + ke^{-ikr_0} \cot kr_0 \right]$$

$$\frac{C}{D} = e^{-2ikr_0} \frac{ik + k \cot kr_0 + \frac{2ma}{\hbar^2}}{ik - k \cot kr_0 - \frac{2ma}{\hbar^2}}$$

$$= e^{-2ikr_0} \left[ \frac{e^{ikr_0} + \frac{2ma}{\hbar^2 k}}{-e^{-ikr_0} - \frac{2ma}{\hbar^2 k}} \right] = -e^{2i\delta_0}$$

$$\text{so } e^{2i\delta_0} = e^{-2ikr_0} \left[ \frac{e^{ikr_0} + \frac{2ma}{\hbar^2 k}}{e^{-ikr_0} + \frac{2ma}{\hbar^2 k}} \right]$$

$$\frac{d\sigma_0}{d\Omega} = |f_0(\theta)|^2 = \left| \frac{1}{k} e^{i\delta_0} \sin \delta_0 \right|^2 = \frac{1}{k^2} \sin^2 \delta_0$$

$$2i \sin 2\delta_0 = e^{-2ikr_0} \left[ \frac{e^{ikr_0} + \frac{2ma}{\hbar^2 k}}{e^{-ikr_0} + \frac{2ma}{\hbar^2 k}} \right] - e^{2ikr_0} \left[ \frac{e^{-ikr_0} + \frac{2ma}{\hbar^2 k}}{e^{ikr_0} + \frac{2ma}{\hbar^2 k}} \right] \quad \text{Define: } \alpha \equiv kr_0; \beta \equiv \frac{2ma}{\hbar^2 k}$$

$$\begin{aligned}
 2i \sin 2\delta_0 &= \frac{e^{-i\kappa} + \beta e^{-2i\kappa}}{e^{-i\kappa} + \beta} - \frac{e^{i\kappa} + \beta e^{2i\kappa}}{e^{i\kappa} + \beta} \\
 &= \frac{(e^{-i\kappa} + \beta e^{-2i\kappa})(e^{i\kappa} + \beta) - (e^{i\kappa} + \beta e^{2i\kappa})(e^{-i\kappa} + \beta)}{(e^{-i\kappa} + \beta)(e^{i\kappa} + \beta)} \\
 &= \frac{(1 + 2\beta e^{-i\kappa} + \beta^2 e^{-2i\kappa}) - (1 + 2\beta e^{i\kappa} + \beta e^{2i\kappa})}{1 + 2\beta \cos \kappa + \beta^2} \\
 &= \frac{-4i\beta \sin \kappa - 2\beta^2 i \sin 2\kappa}{1 + 2\beta \cos \kappa + \beta^2} \quad \text{Take } \kappa \ll 1 \text{ approx:}
 \end{aligned}$$

$$2i \sin 2\delta_0 \cong \frac{-4i\kappa(\beta - \beta^3)}{(1 + \beta)^2} = -4i\kappa \frac{\beta}{1 + \beta}$$

$$\sin 2\delta_0 \approx -2kr_0 \frac{\frac{2ma}{\hbar^2 k}}{1 + \frac{2ma}{\hbar^2 k}}$$

$$\text{so } \sin \delta_0 \approx -kr_0 \frac{\frac{2ma}{\hbar^2 k}}{1 + \frac{2ma}{\hbar^2 k}}$$

$$\frac{d\delta}{d\Omega} = r_0^2 \left( \frac{\frac{2ma}{\hbar^2 k}}{1 + \frac{2ma}{\hbar^2 k}} \right)^2$$

since spherically symmetric,  $\sigma = \int d\Omega \frac{d\sigma}{d\Omega} = 4\pi r_0^2 \left( \frac{\frac{2ma}{\hbar^2 k}}{1 + \frac{2ma}{\hbar^2 k}} \right)^2$

Note that this has correct limiting behavior:

$a \rightarrow 0$ :  $\sigma \rightarrow 0$  since there's no potential

$a \rightarrow \infty$ : looks like hard sphere since the "shell" becomes impenetrable:  $\sigma \rightarrow 4\pi r_0^2$ .